MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6312

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, integral cavity, mechanically tunable 8.5 to 10 GHz, 25 mW minimum power output

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Irs	Tuner plate temperature
Unit:	V	Vdc	Vdc	mAdc	°C
Maximum:	7.0	350	-1,000	42	200
Minimum:	5.7	---	-20	---	---

PHYSICAL CHARACTERISTICS:

See figure 1
Cathode: Unipotential
Mounting position: Any
Cooling: Conduction/Forced air
RF output flange: Mates with UG-39/U
Weight: 8 ounces, approximately (186.62 grams)

TEST CONDITIONS: See note 1

Parameter:	Ef	Ers	Er	F	Tuner plate temperature
Unit:	V	Vdc	Vdc	MHz	°C
Test condition 1:	6.3	300	-55 to -225	F2	100 (max)
Test condition 2:	6.3	300	Mode 5/Max Po	F2	100 (max)

GENERAL:

Qualification - Required

	Frequency	
F	MHz	Tolerance
1	8,500	±0.3%
2	9,300	±0.3%
3	10,000	±0.3%

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:32Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
1031	Operation vibration (1)	-	2	F = 50 to 600 Hz; 10 G; t = 5 minutes	ΔF (p-p)	---	4	MHz
1042	Shock	3	2	Shock duration = 1.0 ms; 200 G	ΔF	---	5	MHz
4250	Power output (3)	-	2	F1 thru F3	Po	25	---	mW
4280	Electronic tuning range (2)	-	2	Er (Mode 5) 1/2 max Po; F1 thru F3	ΔF	30	---	MHz
4027	Temperature coefficient	8	2	TA = 20° to 60°C	Coeff	0.0	-0.20	MHz/°C
4225	Residual FM	-	2	Ef = 5.7 to 7.0 Vdc	ΔF (p-p)	---	0.1	MHz
---	Frequency drift	6	2	t = 10 sec	ΔF	---	±2.0	MHz
---	Heater voltage coefficient	-	1	Ef = 5.7 to 7.0 V	ΔF/ΔEf	---	6	MHz/V
4223	Mechanical tuning rate	-	-	F1 to F3	ΔF/Δ Tuner degrees	1.8	3	MHz/ degrees
	Quality conformance inspection, part 1	9	-					
1031	Operation vibration (2)	2	1	F = 60 Hz; 10 G; t = 2 minutes	Ir	0	10	uAdc
4229	Total reflector current	4	1	t = 2 minutes	Ir	---	3.0	uAdc
4214	Cathode emission	4,10	-	Ef = 5.7 V	ΔIrs/ Irs	---	-15	%
1256	Resonator current	-	2		Irs	20	32	mAdc
4223	Tunable frequency	-	2		F	8,500	10,000	MHz
4250	Power output (1)	-	2		Po	25	---	mW
4213	Reflector voltage (1)	-	2		Er	-115	-180	Vdc
---	Broadband tuning	7	2	σ' = 8 dB; VSWR = 2.5:1	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current	-	-		If	1.08	1.32	A
4250	Power output (2)	-	2	F1 and F3	Po	25	---	mW
4213	Reflector voltage (2)	-	2	F1 and F3	---	---	---	---
					Er	-85	-135	Vdc
					Er	-160	-225	Vdc
4280	Electronic tuning range (1)	-	2	Er (Mode 5) 1/2 max Po	ΔF	30	---	MHz
4231	Hysteresis	5	2		Ratio	---	0.5	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:32Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3							
---	Life test	-	-	Group C	t	500	---	hrs
---	Life-test end points:							
4250	Power output (1)	-	2		Po	20	---	mW
4229	Reflector current	-	1	t = 5 minutes	Ir	---	10	uAdc

NOTES:

1. All oscillation tests except vibration test shall be made with the tube rigidly connected to a UG39/U (MIL-F-3922/1) flange on appropriate RG52/U (MIL-W-85/1) waveguide equipment and the load VSWR for the tube shall be less than 1.1. Forced-air cooling is required for power inputs above 10 watts.
2. The reflector current shall be recorded with a Brush Model BL202 recorder, or equivalent. There shall be no reflector current bursts greater than the limit shown.
3. The tube shall be given five shocks in each of three planes. The frequency shift after shock in any one plane shall not exceed the value specified.
4. The tube shall not be oscillating during the test.
5. Superimposing sufficient 60 Hz ac voltage on the direct reflector voltage to suppress oscillations on the ends of the sweep, the mode curve shall be observed on an oscilloscope whose amplifier has minimum bandpass of 0.1 MHz. The ratio of power output at the point of maximum hysteresis to maximum power output shall not exceed the value specified.
6. The frequency shall be stabilized at a pressure of 70 mmHg. The pressure shall be increased to 760 mmHg and the frequency read within 10 seconds. The resulting frequency change shall not exceed the limit specified.
7. Superimposing sufficient 60 Hz ac voltage on the direct reflector voltage to suppress oscillations on the ends of the sweep, the mode curve shall be observed on an oscilloscope whose amplifier has minimum bandpass of 0.1 MHz (see figure 2). With a standing wave introducer inserted, there shall be no discontinuities at the maximum power point for any phase of the standing wave. If the center of the mode is distorted, a wavemeter pip should not disappear from any interval of the mode center as the wavemeter is tuned to check for a frequency discontinuity.
8. The temperature coefficient, when tested under the conditions specified, shall be computed in frequency units per degree celsius over the range specified.
9. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
10. The resonator current shall not drop more than the specified percentage when the heater voltage is decreased from the normal operating potential to the lower specified value.
11. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

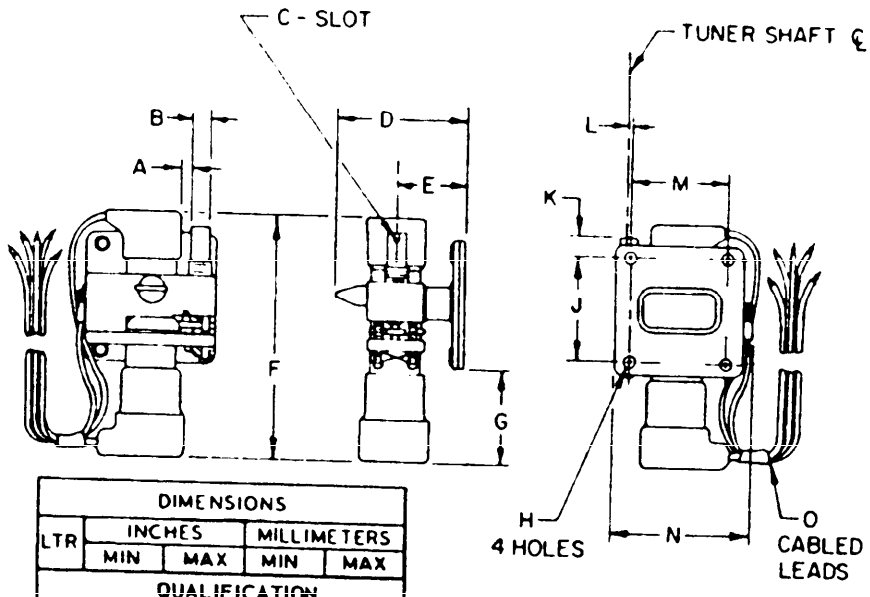
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2877)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:32Z
Check the source to verify that this is the current version before use.



DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALIFICATION				
A	.125 Nom		3.18 Nom	
G		1.000		25.40
K		.260		6.60
N		1.875		47.63
QUALITY CONFORMANCE INSPECTION, PART 1				
D		1.813		46.05
J	1.276	1.284	32.41	32.61
M	1.216	1.224	30.89	31.09
QUALITY CONFORMANCE INSPECTION, PART 2				
B	.248	.250	6.30	6.35
C	.05 wide .22 deep		1.27 wide 5.59 deep	
E	.850	.936	21.59	23.77
F		3.000		76.20
H	.185 Nom		4.70 Nom	
L	.010	.070	.254	1.778
O	18.00 Nom		457.2 Nom	

Cable connections		
Yellow	Heater	
White	Heater	internally connected
Green	Cathode	internally connected
Grey	Reflector	
Brown	Body	

NOTES:

a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

b. Eyelet-type inserts in the flange mounting holes are .219 O.D., .185 I.D., nominal, and are removable from rear of flange. With inserts in place, the mounting holes provide clearance for No. 8 screws. With inserts removed, the mounting holes allow use of insulating bushings where dc isolation is desired.

FIGURE 1. Outline drawing of electron tube type 6312.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:32Z
Check the source to verify that this is the current version before use.

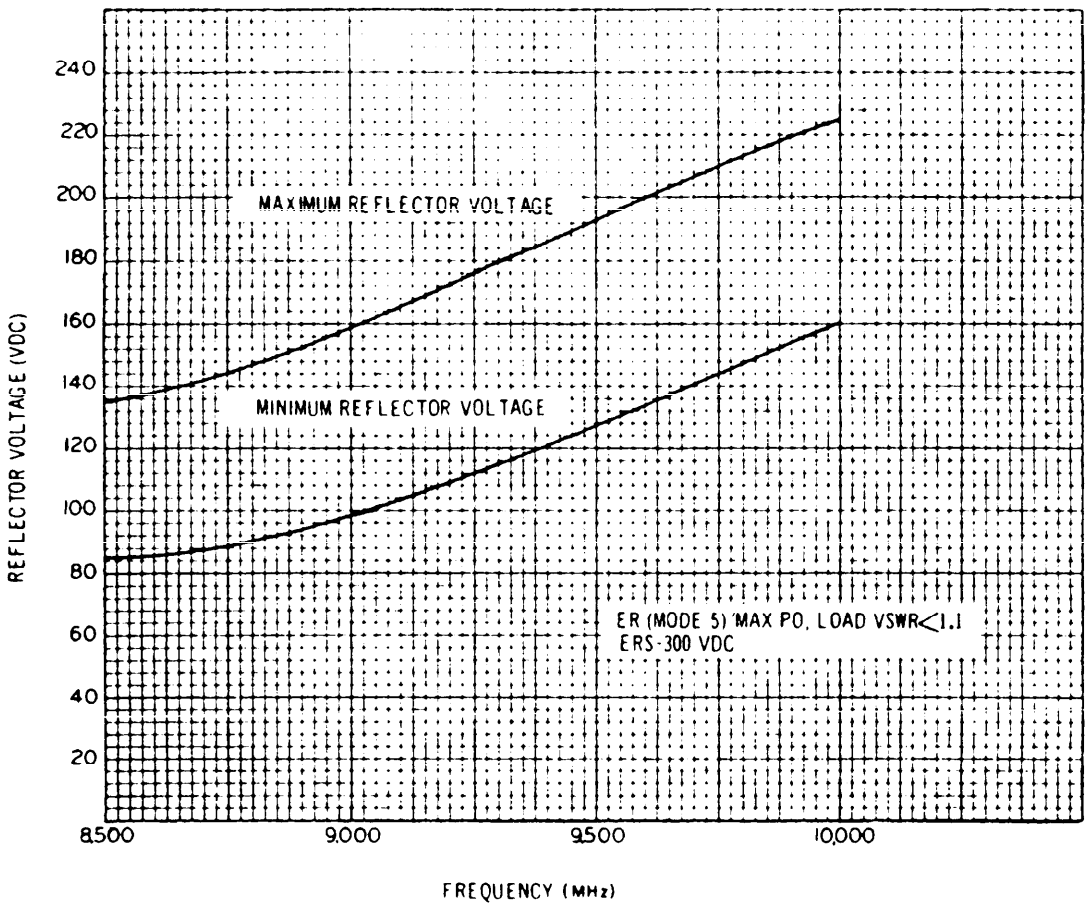


FIGURE 2. Reflector voltage, specifications limits (Vdc) versus frequency (MHz).

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:32Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:32Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:32Z
Check the source to verify that this is the current version before use.